B. Hamon Odyssey

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: bitmasks, greedy, two pointers | https://codeforces.com/problemset/problem/1847/B

Jonathan is fighting against DIO's Vampire minions. There are n of them with strengths a_1, a_2, …, a_n. defand &

Denote (l, r) as the group consisting of the vampires with indices from l to r. Jonathan realizes that the *strength* of any such group is in its weakest link, that is, the bitwise AND. More formally, the *strength* level of the group (l, r) is defined as $f(l,r) = a_l$ and a_{l+1} and a_{l+2} and … and a_r. Here, and denotes the bitwise AND operation.

Because Jonathan would like to defeat the vampire minions fast, he will divide the vampires into contiguous groups, such that each vampire is in **exactly** one group, and the **sum** of *strengths* of the groups is **minimized**. Among all ways to divide the vampires, he would like to find the way with the **maximum** number of groups.

Given the strengths of each of the n vampires, find the **maximum number** of groups among all possible ways to divide the vampires with the smallest sum of *strengths*.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the number of vampires.

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($0 \le a_i \le 10^9$) — the individual strength of each vampire.

The sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum number of groups among all possible ways to divide the vampires with the smallest sum of *strengths*.

Example

Input
3 3 1 2 3 5 2 3 1 5 2 4 5 7 12 6

Output
1 2 1

Note

In the first test case, the optimal way is to take all the n vampires as a group. So, $f(1,3) = 1$ and 2 and $3 = 0$.

In the second test case, the optimal way is to make 2 groups, (2,3,1) and (5,2). So, $f(1,3) + f(4,5) = (2$ and 3 and $1) + (5$ and $2) = 0 + 0 = 0$.